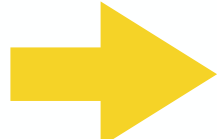


Outline

- General organization of the course
- Object oriented programming
- Basic types and their operations
- Methods

 **UML class diagrams**

Class diagrams: classes

- A graphical visualization of the **Rational** class, considering only the essential functionality, might look like this

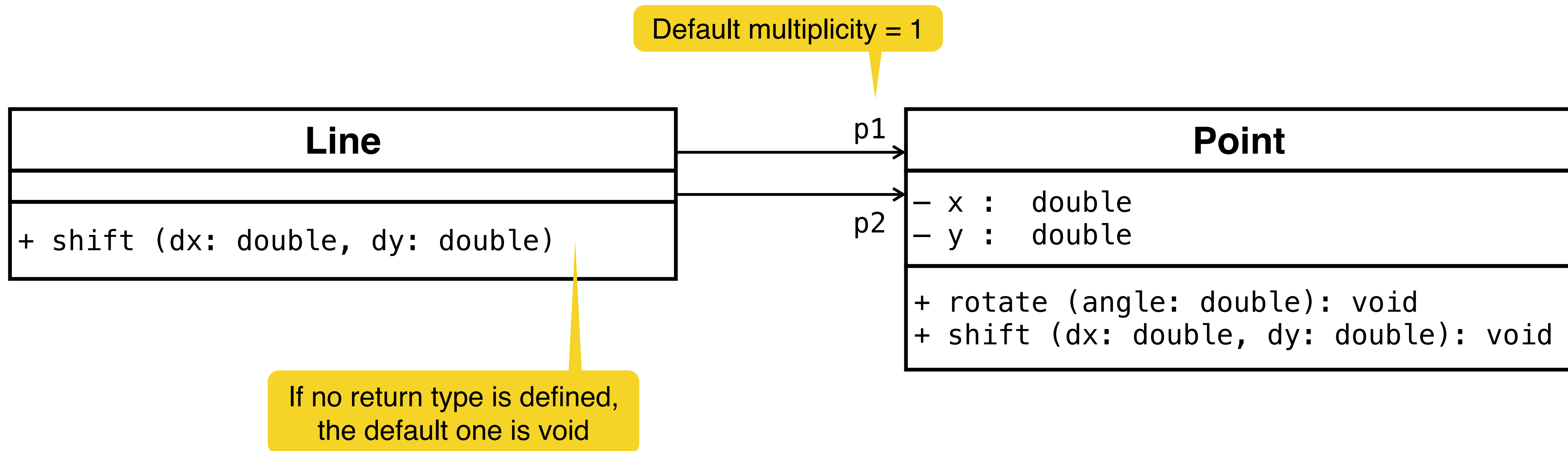
Rational		
–	numerator	: int
–	denominator	: int
+	add	(y: Rational) : Rational
+	equals	(y: Rational) : boolean

Class diagrams: discussion and outlook

- Such diagrams are provided by UML (Unified Modeling Language), to design software systems
 - More on UML in the course
Introduction to Software Engineering
- For a single class, such a diagram is not really worthwhile
- However, if a system consists of many classes, it can be used to illustrate the relationships between different classes
- UML was not developed specifically for Java
 - Types are notated differently
 - Also, some ideas cannot be modeled at all or only poorly

Class diagrams: associations

- Reference attributes are often noted as references (lines between classes)
- In the **example** below, a line is connected to two points p1 and p2



Next steps

- Visit a tutorial group
 - T01E01 - Calculator
 - T01E02 - Autumn Leaves
- Solve the homework independently
 - H01E01 - Lectures
 - H01E02 - Take-Off in Munich
- Read the following articles to find out more about object oriented programming in Java
 - https://www.w3schools.com/java/java_oop.asp
 - https://www.w3schools.com/java/java_methods.asp

Click on **Next** several times

→ Due until **Wednesday, October 23, 13:00**

- **Object oriented programming** is the elementary thinking in Java and many other programming languages
 - **Attributes** allow to model the state of objects
 - **Constructors** initialize objects
 - **Methods** implement the logic (behavior) of objects
- Eight **basic data types**: boolean, byte, short, int, long, float, double, char
- **Methods** determine the behavior of objects
 - Static methods belong to classes
- **UML class diagrams** allow to visualize the structure and relationships of classes

References

- <https://www.javatpoint.com/difference-between-object-and-class>
- <https://www.javatpoint.com/object-and-class-in-java>
- <https://www.javatpoint.com/method-in-java>
- <https://www.javatpoint.com/java-constructor>
- <https://www.javatpoint.com/static-keyword-in-java>
- <https://www.javatpoint.com/this-keyword>
- <https://www.javatpoint.com/java-oops-concepts>
- <https://www.geeksforgeeks.org/difference-between-class-and-object>